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Department of CSE

B.Tech. CSE Trimester -5

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Cloud Infrastructure and Services

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LAB RECORD

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Experiment -2

AIM: Enable MFA on the root user of your account and prepare a short note comparing virtual MFA devices, hardware MFA devices and FIDO security keys.

Description:

YubiKey is a physical FIDO security key used for multi-factor authentication (MFA). It helps verify a user's identity while logging into online accounts and services.

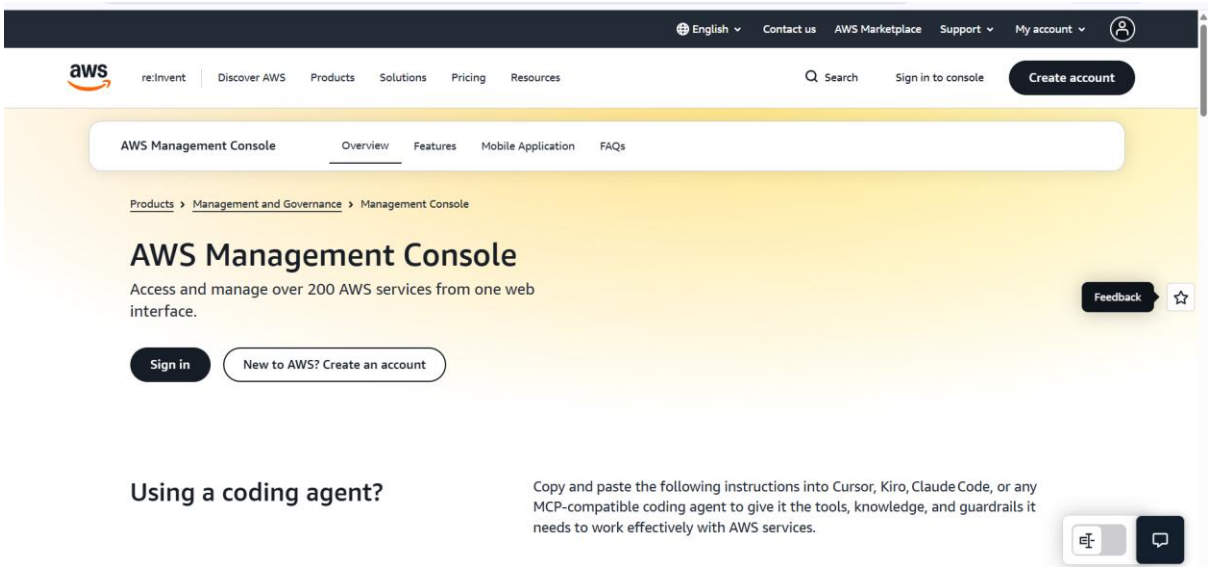
It connects to devices through USB or NFC and provides secure authentication without requiring users to enter six-digit OTP codes. It is easy to use and offers strong protection against phishing attacks.

YubiKey is commonly used to secure AWS accounts and other online services. It provides an additional layer of security by preventing unauthorized users from accessing an account, even if they know the password.

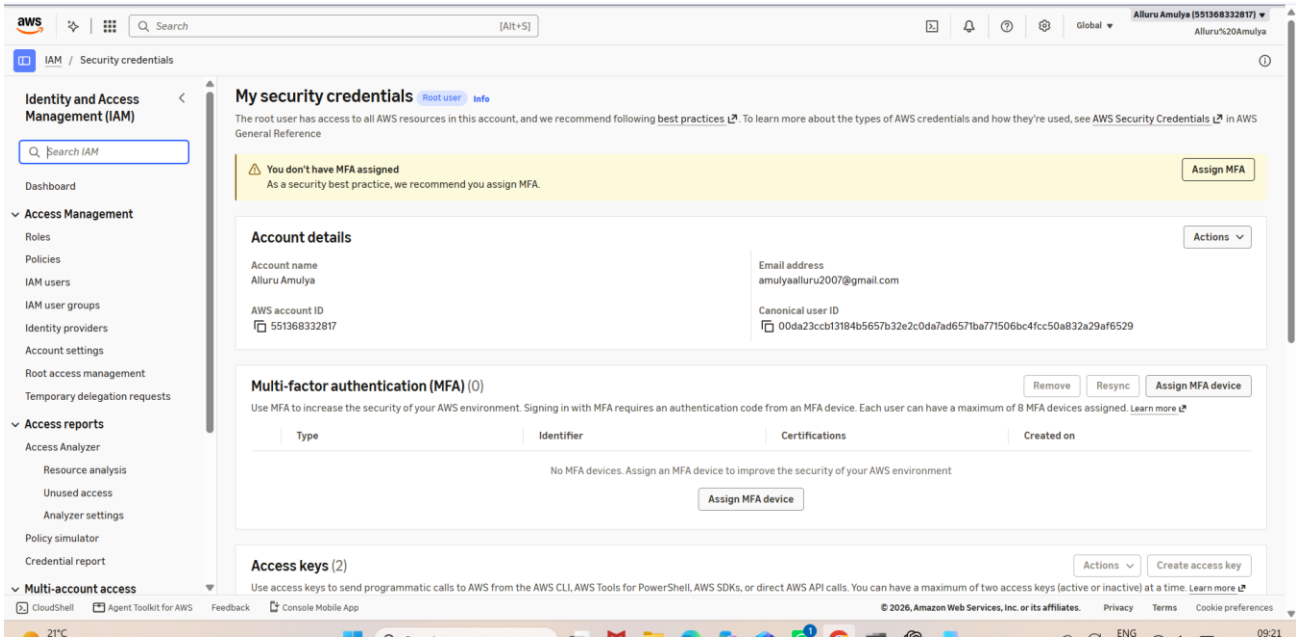
Steps:

Part A: Enable MFA on the AWS Root User

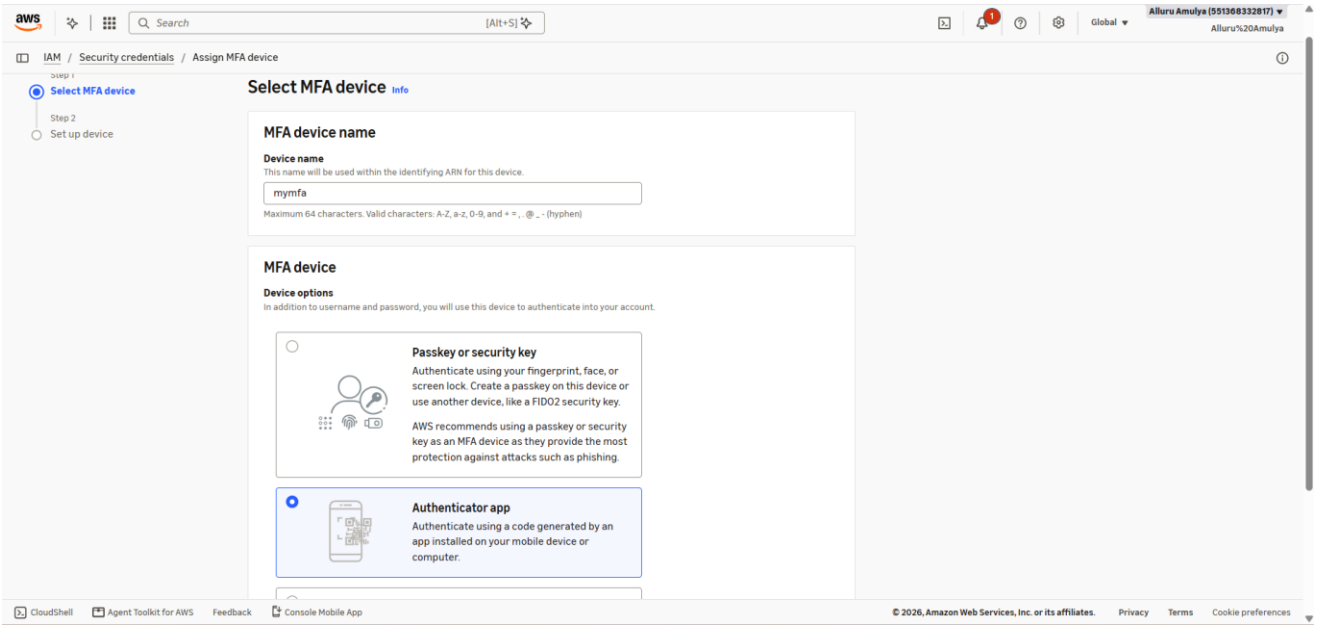
1. Open the AWS Management Console by visiting https://aws.amazon.com/console/.
2. Sign in using the root user email address and password associated with the AWS account.



3. Complete any verification steps requested by AWS.
4. In the AWS Management Console, click the account name or account menu in the top-right corner.
5. Select Security credentials.

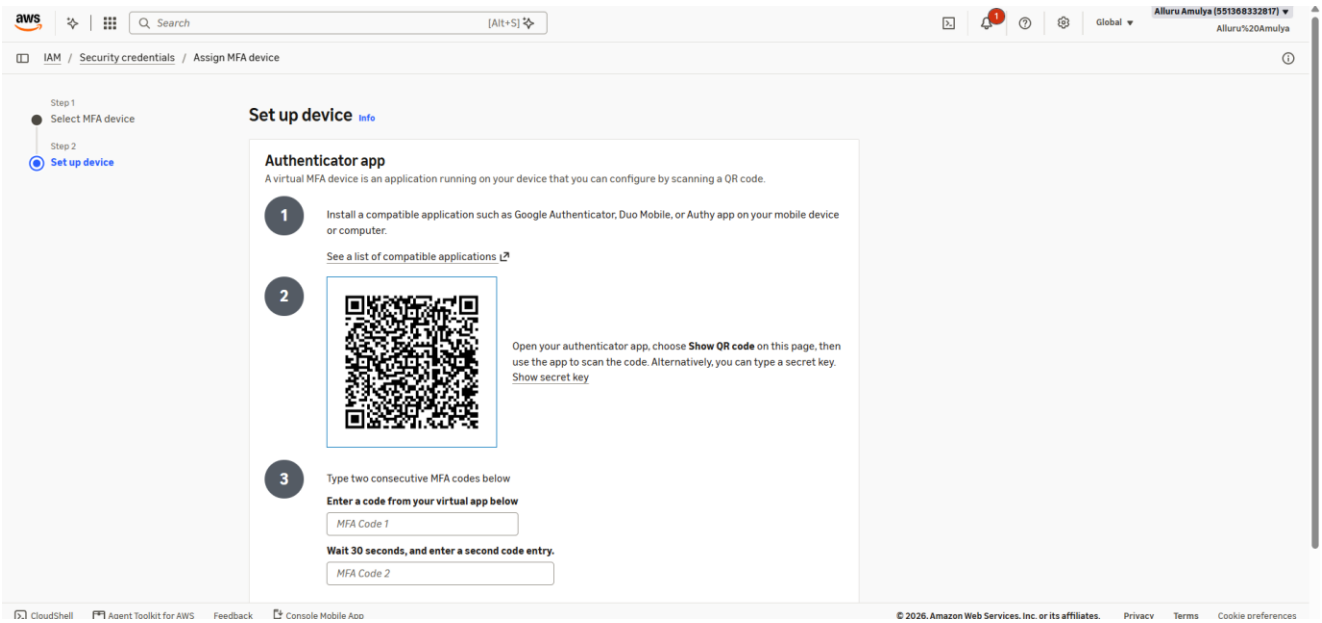


6. Locate the Multi-factor authentication (MFA) section.
7. Click Assign MFA device.
8. Enter a suitable device name (e.g., Root-User-MFA).
9. Select the MFA device type. For this experiment, select Authenticator app (Virtual MFA device).



10. Click Next.

11. AWS displays a QR code on the screen.

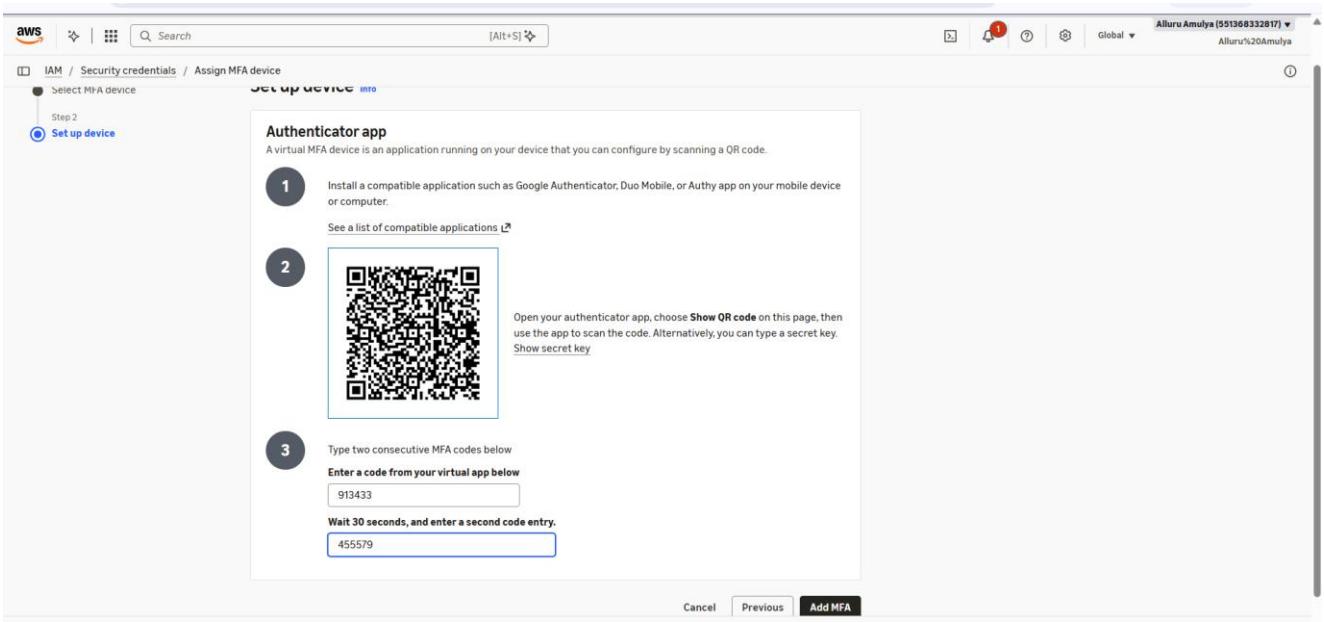


12. Open an authenticator application on your smartphone, such as Google Authenticator or Microsoft Authenticator.

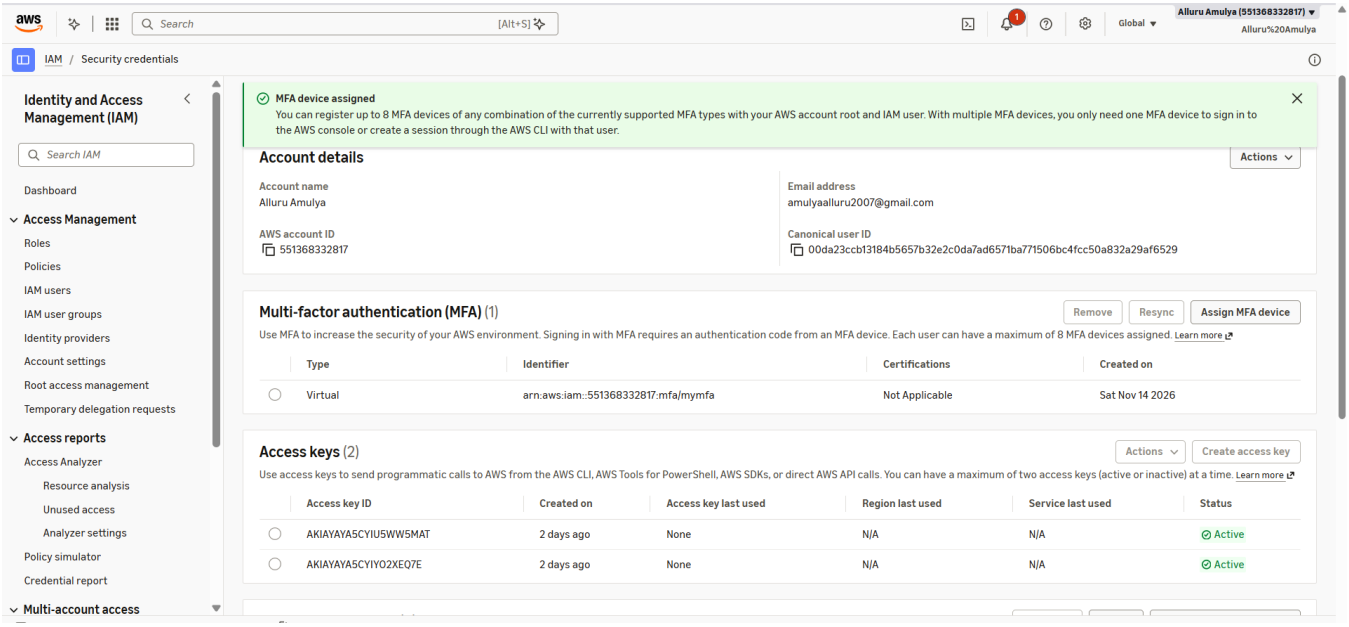
13. Select the option to add a new account or MFA device.

14. Scan the QR code displayed on the AWS console using the authenticator application.

15. The application starts generating a six-digit time-based one-time password (TOTP).
16. Enter the first six-digit code in the MFA code 1 field.
17. Wait for the application to generate the next six-digit code.
18. Enter the second six-digit code in the MFA code 2 field.

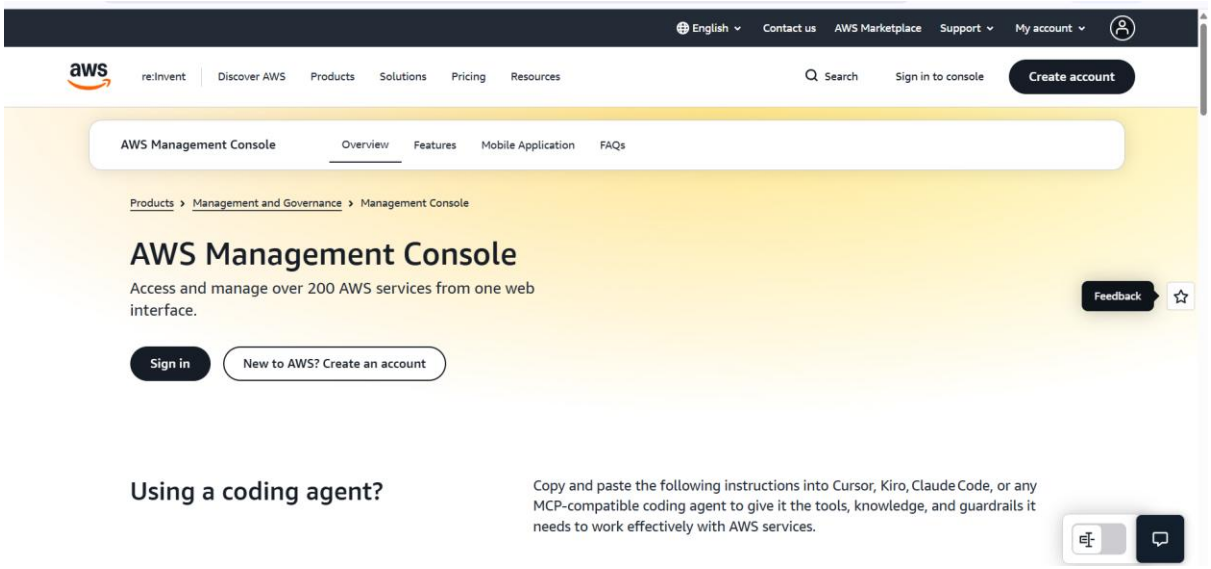


19. Click Add MFA to complete the registration.
20. Verify that the virtual MFA device is displayed under the root user's Multi-Factor Authentication (MFA) section.

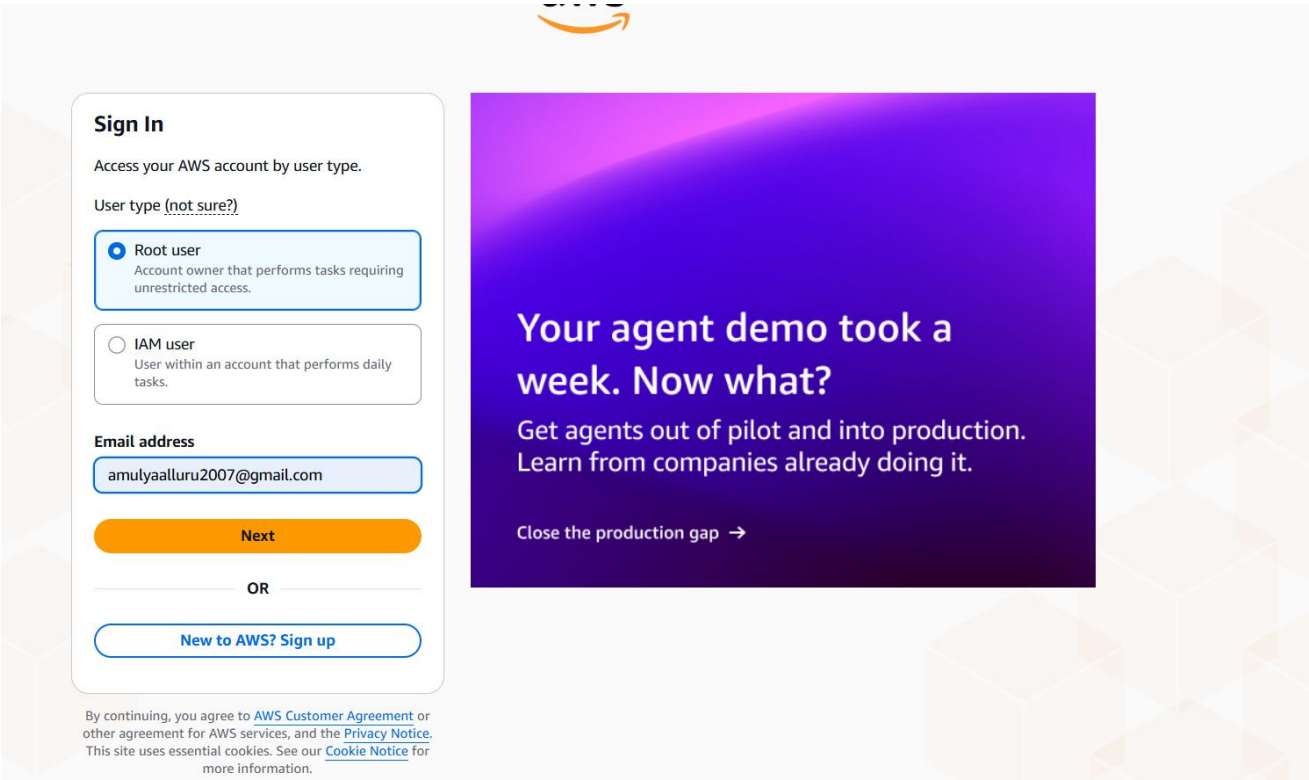


Part B: Verify MFA Authentication

1. Sign out of the AWS Management Console.
2. Open the AWS Management Console sign-in page.

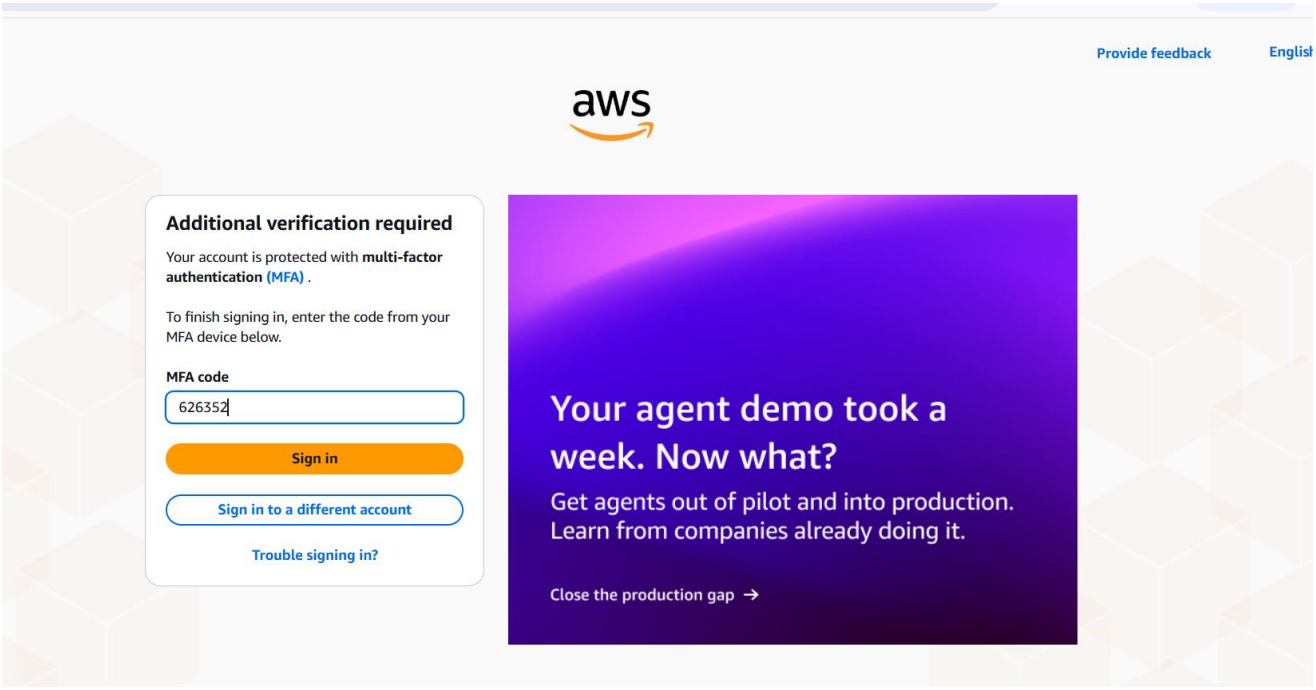


3. Select Root user.
4. Enter the registered root user email address and password.

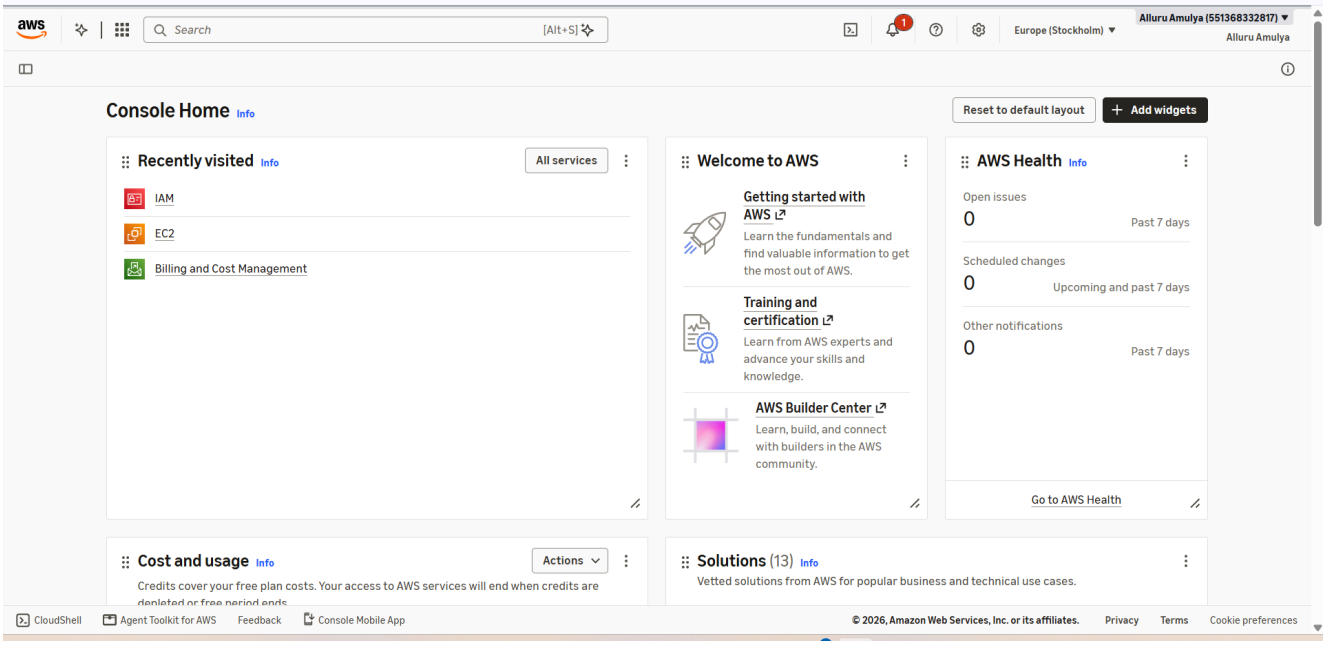


5. AWS prompts for the MFA authentication code.

6. Open the authenticator application on your smartphone.
7. Enter the currently displayed six-digit authentication code.

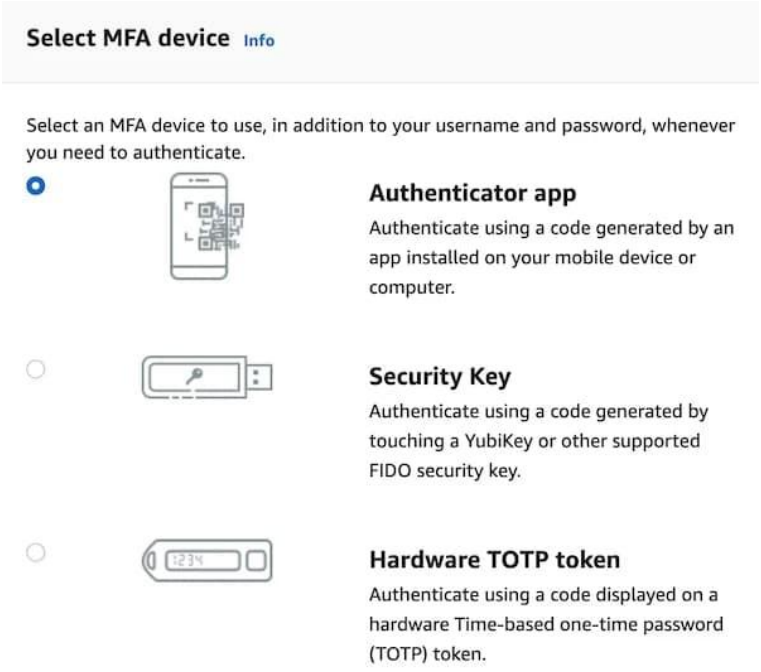


8. Click Submit or the equivalent verification option.
9. Verify that you can successfully access the AWS Management Console.



Note: The MFA code changes periodically. Enter the current code before it expires.

Part C: Prepare a detailed comparative Note on Different MFA (Multi Factor Authentication) Devices- Virtual MFA devices, hardware MFA devices and FIDO (Fast IDentity Online) security keys.



Virtual MFA devices – Google Authenticator, Microsoft Authenticator

Hardware MFA devices – A hardware MFA device is a dedicated physical token that generates a time-based, six-digit authentication code. The user enters this code during AWS sign-in.

Example: AWS-supported hardware TOTP security tokens.

FIDO Security Keys -- are physical devices used to verify a user's identity during login.

Examples: YubiKey and other AWS-supported FIDO security keys.



YubiKey

Output:

